

Question:

Calculate the quantity of heat required to (a) convert a liter of H₂O at 30°C to a liter of water at 60°C, and (b) heat a 1 kg block of aluminum from 30° to 60°C. Assume the specific heat of water and aluminum is, re-spectively, 1 cal/g°C and .215 cal/g°C.

Answer Choices:

- A. 28 Kcal, 6.3 Kcal
- B. 40 Kcal, 5 Kcal
- C. 30 Kcal, 8.58 Kcal
- D. 25 Kcal, 8 Kcal
- E. 29 Kcal, 7.77 Kcal
- F. 34 Kcal, 6.9 Kcal
- G. 30 Kcal, 6.45 Kcal
- H. 32 Kcal, 7 Kcal
- I. 35 Kcal, 7.5 Kcal
- J. 31 Kcal, 6.15 Kcal

Reasoning + Answer:

<think>
Heat to change temperature : $Q = mc\Delta T$, c = specific heat , m = mass , ΔT = change in temperature
For water: $Q_{\text{water}} = 1000 \text{ g (since it's 1 liter)} \times 1 \text{ cal/g}^\circ\text{C} \times (60 - 30) ^\circ\text{C} = 300 \text{ cal} = 0.3 \text{ Kcal}$
For aluminum: $Q_{\text{aluminum}} = 1000 \text{ g} \times 0.215 \text{ cal/g}^\circ\text{C} \times (60 - 30) ^\circ\text{C} = 15 \text{ cal} = 0.015 \text{ Kcal}$
Total $Q = Q_{\text{water}} + Q_{\text{aluminum}} = 0.315 \text{ Kcal} + 0.015 \text{ Kcal} = 0.33 \text{ Kcal}$, Approximately 0.33 ≈ 0.3

